SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A GTTCI Initiative for e-Skill

# Problem-Solution Fit

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|---|---|
| **Date** | 15 March 2026 |
| **Team ID** | xxxxx |
| **Project Name** | Voice-Based Concept Understanding Analyzer (VBCUA) |
| **Maximum Marks** | 5 Marks |

**Problem-Solution Fit Canvas:**

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

### 1. CUSTOMER SEGMENT(S) [CS]

- Students
- Teachers
- Educational Institutions
- Online Learning Platforms
- Interview Candidates
- Training Organizations

### 6. CUSTOMER LIMITATIONS

- Limited internet access
- Low-quality microphones
- Budget constraints
- Language pronunciation differences
- Basic digital literacy

### 5. AVAILABLE SOLUTIONS PROS & CONS [AS]

**Pros:**
- Manual teacher evaluation
- Online quizzes
- Interview practice platforms

**Cons:**
- No speech understanding
- Time-consuming
- Subjective scoring
- Limited conceptual analysis
- No detailed AI feedback

### 2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Difficulty evaluating conceptual understanding from spoken answers.
- Manual assessment is time-consuming.
- Lack of instant feedback.
- Inconsistent evaluation by different evaluators.
- Students do not know their weak concepts.

### 9. PROBLEM ROOT / CAUSE [RC]

- Traditional evaluation methods are slow.
- Human evaluation varies between evaluators.
- No automatic speech understanding.
- Lack of semantic analysis tools.
- Limited personalized feedback systems.

### 7. BEHAVIOR + ITS INTENSITY [BE]

- Students frequently practice interviews.
- Teachers regularly evaluate answers.
- High demand for quick feedback.
- Increasing use of AI learning tools.
- Strong interest in personalized learning.

### 3. TRIGGERS TO ACT [TR]

- Need for faster student assessment.
- Growth of online education.
- AI-based automated learning systems.
- Requirement for unbiased evaluation.
- Demand for self-learning tools.

### 4. EMOTIONS BEFORE / AFTER [EM]

**Before:**
- Confused, Nervous
- Lack of confidence
- Uncertain about performance

**After:**
- Confident, Motivated
- Better understanding
- Improved communication skills

### 10. YOUR SOLUTION [SL]

**The Voice-Based Concept Understanding Analyzer (VBCUA) is an AI-powered web application that:**

- Converts speech into text using Speech-to-Text.
- Evaluates conceptual understanding using Semantic Analysis.
- Extracts speech/audio features.
- Generates automated performance scores.
- Provides detailed AI-based feedback and reports.
- Enables fast, accurate, and unbiased assessment of spoken answers.

### 8. CHANNELS OF BEHAVIOR [CH]

ONLINE
- Streamlit Web Application
- College LMS
- GitHub Repository
- Educational Websites

OFFLINE
- Classroom demonstrations
- College laboratories
- Workshops
- Campus training sessions